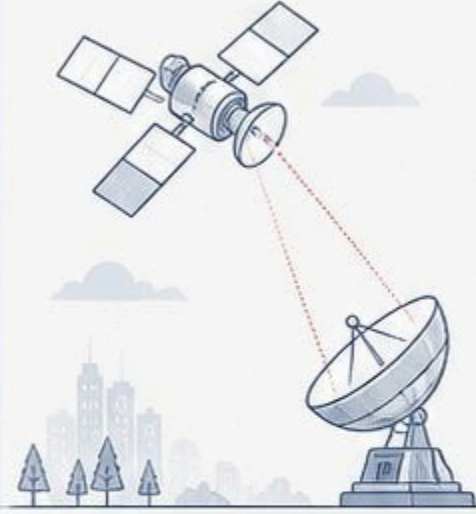


1 ABSTRACT



Q-Orbit is a theoretical research project that evaluates a satellite quantum key distribution (QKD) workflow under controlled device-imperfection assumptions. The study cross-checks locked computational artifacts, regression tests, and audit records to identify what may be claimed safely and what must remain fail-closed.

2 METHODOLOGY

- Controlled JSON and CSV artifact verification
- 12/12 regression tests passed
- 20/20 audit checks passed
- 41 × 41 coupled screen analyzed (1,681 grid points)
- Integer window optimization from 1 to 221 s
- Local sensitivity and proof-to-device mapping reviewed



KEY RELEASE:
QUARANTINED



ZERO RELEASE:
ENFORCED



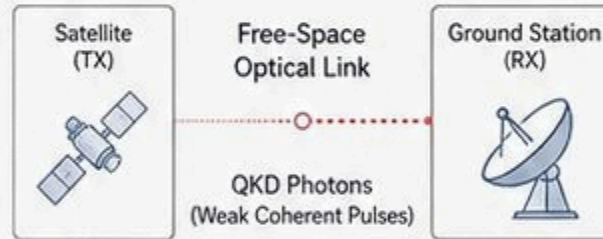
THEORETICAL / NOT PHYSICALLY VALIDATED

3 OBJECTIVE



To verify finite-key performance, optimize the operating window, analyze sensitivity to critical parameters, classify device-imperfection effects, and preserve strict claim-control boundaries for a theoretically defensible Phase 1 result.

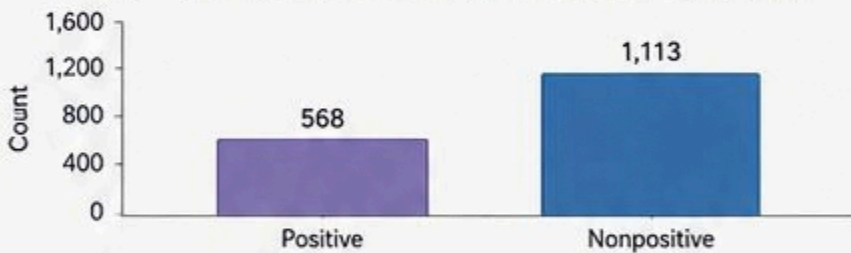
4 SYSTEM DESIGN (THEORETICAL FRAMEWORK)



- Quantum channel: free-space optical link
- Protocol basis: BB84
- Link budget and channel models: theoretical assumptions
- Security framework: finite-key analysis

5 DATA HIGHLIGHT

41 × 41 Coupled Screen Classification (1,681 Grid Points)

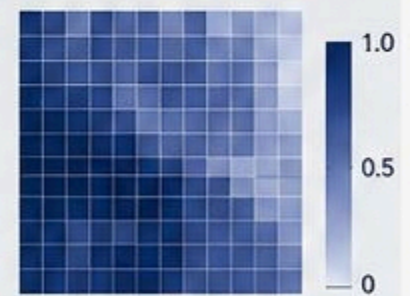


	Grid Points: 1,681
	Grid Median Signed Margin: -2,624.946810258186 bits
	Grid Minimum Signed Margin: -3,828.414517626367 bits

6 KEY INSIGHTS

	Candidate Key	41,338 bits
	Optimized Half-Window	102 s
	Regression Status	12/12 PASS
	Audit Status	20/20 PASS

LOCAL SENSITIVITY (PROOF-TO-DEVICE)



41 × 41 Coupled Screen (Local Sensitivity Map)

7 CONCLUSION



Phase 1 is complete with **CONDITIONAL PASS – READY FOR PHASE 2 WITH DISCLOSED LIMITATIONS.**

Theoretical results support continued development, but physical validation has not yet occurred. Prototype and website are still in progress.



Theoretical Foundation Established



Controlled Verification Completed



Phase 2 Prepared

8 IMPLEMENTATION TOOLS



Python



Controlled CSV/JSON



Local Dashboard



Markdown/PDF Reporting

All results derived from controlled theoretical analysis.
No physical deployment or live-key release.